

# The `mousik` package\*

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## Abstract

This is a music-related package focused on writing music fragments. It works specially well with small examples, such as excerpts for theoretical articles. It is easier than other music writing methods in  $\text{\LaTeX}$  because it is based on the `tikz` package.

## Keywords

*mousik, music, music theory, music notation, music writing, score, sheet, tikz diagram, bar, clef, time signature, note, symbol, rest, dot, chord, accidental, sharp, flat, natural, key*

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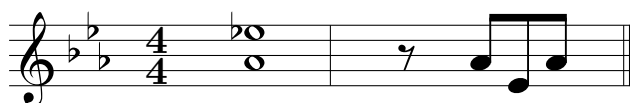
## 1 Introduction

There are already several ways to set music using Tex. As stated by the `abc` package, “*notably MusicTEX and Lilypond*”, as well as `abc per se`. Some of them are too difficult to use for simple music; others do not work at all.

The `mousik` package is perfect for short, one line music fragments. It may be used for longer scores, but there are more adaptable packages than this one. However, in spite of its compactness, the package is quite complete and versatile, but easy to learn and use.

## 2 The `mousik` environment

The way to draw a score is by means of the `mousik` environment.



```

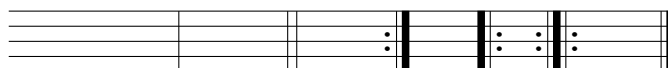
\begin{mousik}
  \Clef{2}
  \Key{-3}
  \TimeSig{4}{4}

  \Note{1}{5}
  \Chord
  \Acc{-1}{9}
  \Note{1}{9}
  \Bar

  \Rest{8}
  \PolyNote{8}{5,2,5}
  \FinalBar
\end{mousik}

```

## 2.1 Ending a line: bars



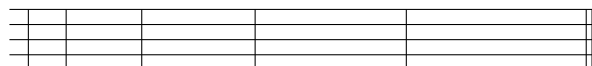
the bar command

|                     |                   |                         |                                                     |                        |
|---------------------|-------------------|-------------------------|-----------------------------------------------------|------------------------|
| <code>\Space</code> | <code>\Bar</code> | <code>\DoubleBar</code> | <code>\RepeatBar</code>                             | <code>\FinalBar</code> |
|                     |                   |                         | <code>{L}</code> <code>{B}</code> <code>{LB}</code> |                        |

The only way to end a line in the score is with a bar. Any other way may cause visual errors (see subsection 4.1).

## 2.2 The \Space command

The `\Space` command is not only used when no bar is needed. It may also be used to input *extra* space between symbols. The default space is 2 (which is the default space between symbols in general), but it may be any whole number (see subsection 4.2).

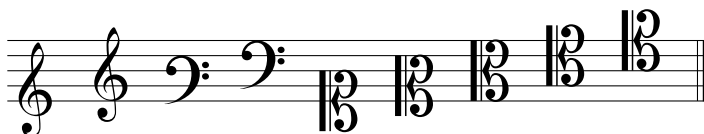

$$\begin{array}{cccc}\backslash\mathrm{Space}[-1] & \backslash\mathrm{Space}[1] & \backslash\mathrm{Space} & \backslash\mathrm{Space}[3] \\ & \backslash\mathrm{Space}[0] & & \end{array}$$

### 3 Common symbols

### 3.1 The `\Clef` command

Clefs might be used at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

- `\Clef{1}` and `\Clef{2}` are Treble Clefs in the first and second lines.
- `\Clef{3}` and `\Clef{4}` are Bass Clefs in the third and fourth lines.
- `\Clef{11}`, `\Clef{12}`, `\Clef{13}`, `\Clef{14}` and `\Clef{15}` are Alto Clefs in each of the five lines.

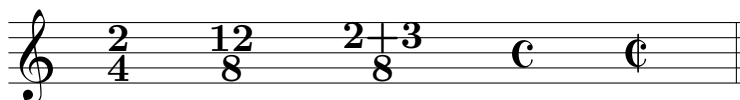


## 3.2 The `\TimeS` command

Time signatures might be used at any moment on the score. `\TimeSig{2}{4}` will draw a 2 by 4 time signature. Any number or symbol can be written inside the brackets.

Also `\TimeSig[1]{}{}` will produce a *common time* symbol and `\TimeSig[2]{}{}` will produce an *alla breve* symbol.

Clean up  
alla breve

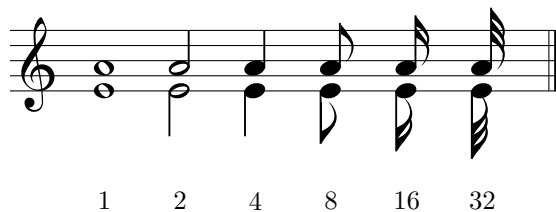


## 3.3 Notes and rests

### 3.3.1 The `\Note` command

The `\Note` command prints a note on the score. It accepts 2 arguments.

The first argument is the duration of the note, from one of these: 1 (whole note / semibreve), 2 (half note / minim), 4 (quarter note / crotchet), 8 (eighth note / quaver), 16 (sixteenth note / semiquaver), 32 (thirty second note / demisemi-quaver).



The second argument is the absolute height. For example, these are the notes from 0 to 9:

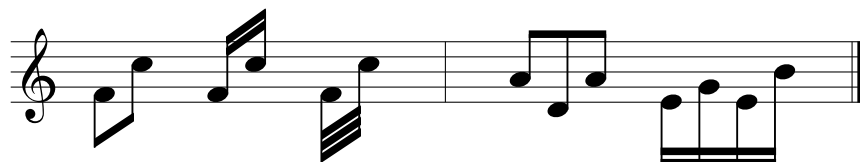


### 3.3.2 The `\PolyNote` command

fuse note  
and poly-  
note to-  
gether

The `\PolyNote` command prints several notes tied together by a beam. It accepts 2 arguments: the duration of the note and the list of heights separated by commas: `\PolyNote{8}{6,3,6}`.

When there are 2 notes, the beam is diagonal; when there are more notes, the beam becomes horizontal.



monotone triplets sliding beam

The notes (or polynotes) go upside down when their height (or maximum height) is bigger than 6. However, for manual control, the `\Swap` command might be used before them, as seen above in the second group.

The `\PolyNote` command might accept a list of durations as the first argument for a mixed beam.

ta taca beams

numerito del tresillo

### 3.3.3 The `\Rest` command

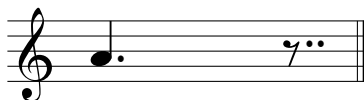
The `\Rest` command prints a rest. It accepts its duration as an argument: 1, 2, 4, 8, 16 or 32.



### 3.3.4 The `\Dot` command

The `\Dot` command prints any amount of dots in front of a note or rest. It accepts an optional argument and a compulsory argument.

The default number of dots is 1, but it may be any whole number bigger than one. The second, compulsory argument is the height of the dot: `\Dot[2]{7}`.



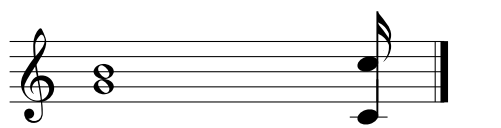
`\Dot{5}`      `\Dot[2]{7}`  
`\Note{4}{5}`      `\Rest{8}`

### 3.3.5 The `\Chord` command

Chords may be obtained by means of the `\Chord` command. It accepts an optional argument: the number of notes to step backwards —mostly for the `\PolyNote` command— with 1 as default.

The best way to use it is to call the `\Note` or `\PolyNote` commands with the highest notes; then call the `\Chord` command; then call the `\Note` or `\PolyNote` commands with the duration 4 of a quarter note, to print them on top.

If the stem needs to be longer, the user may need the `\Enlarge` command, which accepts as arguments the 2 heights of both notes.



```

\Note{1}{6}      \Swap\Note{16}{7}
  \Chord          \Chord
\Note{1}{4}      \Swap\Enlarge{7}{0}
                  \Swap\Note{4}{0}

```

For extreme uses of the command, extra `\Space` commands might be needed, as seen here:



```

\PolyNote{8}{5,4} \PolyNote{8}{5,4} \PolyNote{8}{5,4}
  \Chord[2]        \Chord          \Chord[2]
\PolyNote{4}{1,2} \Note{4}{2}      \Note{4}{1}
                                   \Space[1]

```

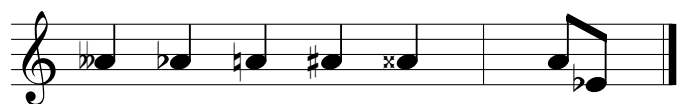
### 3.4 The `\Acc` command

The `\Acc` command prints an accidental before any note. It accepts 2 arguments.

The first argument is the type of accidental:

- -2 for the  $\flat\flat$  double flat symbol.
- -1 for the  $\flat$  flat symbol.
- 0 for the  $\natural$  natural symbol.
- 1 for the  $\sharp$  sharp symbol.
- 2 for the  $\sharp\sharp$  double sharp symbol.

The second argument is the height of the symbol. For usage related to the `\PolyNote`, the `\Space` command is still useful.

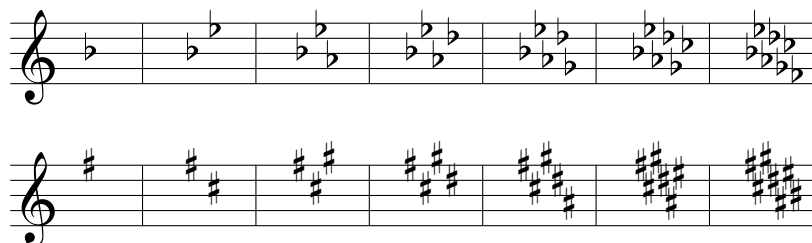


`-2   -1   0   1   2   \Space[1]`  
`\Acc{-1}{2}`  
`\Space[-1]`  
`\PolyNote{8}{6,2}`

### 3.5 The `\Key` command

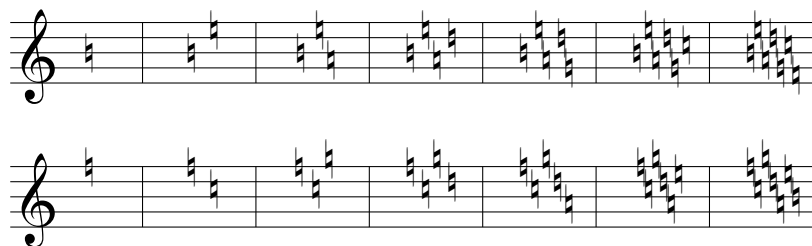
The `\Key` command prints the key at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

It accepts an argument: a positive whole number from 1 to 7 for sharps  $\sharp$ , and a negative whole number from -1 to -7 for flats  $\flat$ .



### 3.6 The `\NoKey` command

The `\NoKey` command is the same as the `\Key` command, but it prints naturals  $\natural$  instead of sharps or flats.



Be careful: the `\Key` and `\NoKey` commands are affected by the current clef! Nonetheless, other commands such as `\Note` or `\Acc` are not affected and need absolute height values.



```

\begin{mousik}
  \Clef{4}
  \Key{1}
  \TimeSig[2]{}{}

  \Acc{0}{8}
  \Note{1}{8}
  \FinalBar
\end{mousik>

```

### 3.7 The `\Text` command

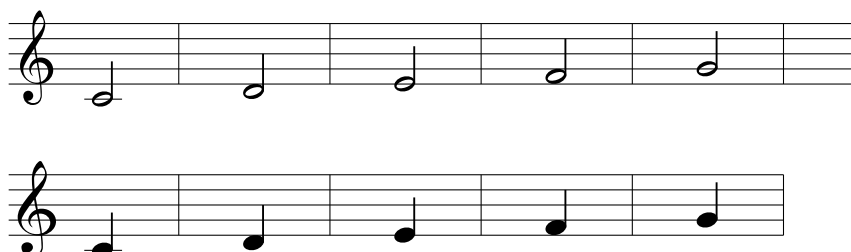
The `\Text` command allows the printing of any text string, including dynamics and lyrics. It has a compulsory argument, but also an optional one where the user may choose the height of the string: `\Text[2]{\it rit.}`.



explain lig-  
aduras

## 4 Observations

### 4.1 Overflow of lines





## 4.2 How do spaces work?

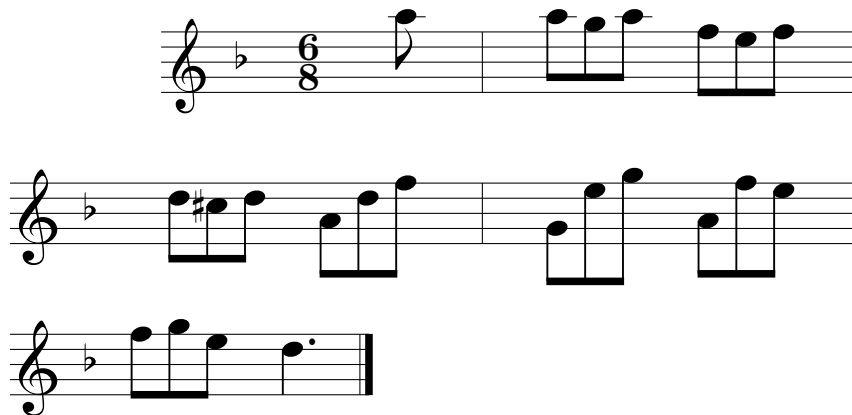
Spaces serve as the manual progression of an invisible cursor during the printing.  
A negative

explain  
spaces

## 4.3 title

Be careful not to use any package that oversteps the alto bla bla such as musixtex `\ClefMX` vs `\Clef`

## 5 A very long and complete example



explain in-  
dent

## 6 The package code

aumentar  
longitud

```
1 \RequirePackage[minimal]{leadsheets}
  \useleadsheetslibraries {musicsymbols}
3 \RequirePackage{harmony}

5 \RequirePackage{tikz}

7 %\usepackage{etoolbox}

9 %\newcommand{\mytest}[1]{%
  % \ifboolexpr{ test {\ifnumcomp{#1}{=}{1}} or test {\ifnumcomp{#1}{=}{14}} }
11 % {do this}
  % {do that}}
13
```

beautiful  
code

```

15  %%%%%%%%%%%
17  \newcounter{Mevaluate}

19  \newcounter{Mmodulo}
    \newcommand{\MModulo}[2]{
21      \setcounter{Mmodulo}{#1}
        \loop
23      \setcounter{Mevaluate}{#2}\addtocounter{Mevaluate}{-1}
        \ifnum\theMmodulo>\theMevaluate
25          \addtocounter{Mmodulo}{-#2}
            \repeat
27      \ifnum\theMmodulo<0
          \addtocounter{Mmodulo}{#2}
29      \repeat
    }

31  \newcounter{Mmax}
33  \newcommand{\MMax}[2][\theMmax]{
    \setcounter{Mmax}{#1}
35    \ifnum#2>#1
      \setcounter{Mmax}{#2}
37    \fi
  }

39  \newcounter{Mmin}
41  \newcommand{\MMin}[2][\theMmin]{
    \setcounter{Mmin}{#1}
43    \ifnum#2<#1
      \setcounter{Mmin}{#2}
45    \fi
  }

47  %%%%%%%%%%%
49  \newcounter{Mfirst}
51  \newcounter{Mxpos}
    \newcounter{Mypos}
53  \newcounter{Mclef}
    \newcounter{Mkey}
55  \newcounter{Mindent}

57  \newenvironment{mousik}[1][0]
    {\setcounter{Mxpos}{0}
59     \setcounter{Mypos}{0}
        \setcounter{Mclef}{0}
61     \setcounter{Mkey}{0}
        \setcounter{Mindent}{#1}
63     \noindent\begin{tikzpicture}\Space[#1]}
    {\addtocounter{Mxpos}{-2}\MStaff
65     \setcounter{Mxpos}{0}
        \setcounter{Mypos}{0}
67     \setcounter{Mclef}{0}
        \setcounter{Mkey}{0}
69     \setcounter{Mindent}{0}
        \end{tikzpicture}}
71

```

```

\newif\ifswap
73 \swapfalse
\newif\ifnoswap
75 \noswaptrue
\newcommand{\Mswap}{
77 \ifnoswap\noswapfalse
\else\noswaptrue
79 \fi
}
81 \newcommand{\Swap}{
\ifswap\swapfalse
83 \else\swaptrue
\fi
85 }

87 \newcommand{\MStaff}{
\foreach\i in {0,...,\theMypos} {
89 \foreach\j in {0,...,4} {
\ifnum\i=\theMypos
91 \ifnum\i=0
\draw (-0.25+\theMindent*0.5,-\i*2+\j*0.2) -- (\theMxpos*0.5,-\i*2+\j*0.2);
93 \else
\draw (-0.25,-\i*2+\j*0.2) -- (\theMxpos*0.5,-\i*2+\j*0.2);
95 \fi
\else\ifnum\i=0
97 \draw (-0.25+\theMindent*0.5,-\i*2+\j*0.2) -- (11,-\i*2+\j*0.2);
\else
99 \draw (-0.25,-\i*2+\j*0.2) -- (11,-\i*2+\j*0.2);
\fi\fi
101 }
}
103 }

105 \newcommand{\Clef}[1]{
\setcounter{Mclef}{#1}
107 \Adv

109 \ifnum#1=1
\node at (\theMxpos*0.5+0.1,-\theMypos*2+0.15)
111 {\resizebox{0.65cm}{0.8cm}{\trebleclef}};
\fi\ifnum#1=2
113 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.35)
{\resizebox{0.65cm}{0.8cm}{\trebleclef}};
\fi\ifnum#1=3
115 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.42)
{\resizebox{0.8cm}{0.6cm}{\bassclef}};
\fi\ifnum#1=4
119 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.62)
{\resizebox{0.8cm}{0.6cm}{\bassclef}};
\fi\ifnum#1=11
121 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.42)
{\resizebox{0.65cm}{0.55cm}{\altoclef}};
\fi\ifnum#1=12
123 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.62)
{\resizebox{0.65cm}{0.55cm}{\altoclef}};
\fi\ifnum#1=13
125 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.82)
127 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.82)

```

```

129     {{\resizebox{0.65cm}{0.55cm}{\altoclef}}};
      \fi \ifnum#1=14
131     \node at (\theMxpos*0.5+0.1,-\theMypos*2+1.02)
      {{\resizebox{0.65cm}{0.55cm}{\altoclef}}};
133     \fi \ifnum#1=15
      \node at (\theMxpos*0.5+0.1,-\theMypos*2+1.22)
135     {{\resizebox{0.65cm}{0.55cm}{\altoclef}}};
      \fi
137     \addtocounter{Mxpos}{2}
139   }

\newcommand{\ClefMX}[1]{
141   \setcounter{Mclef}{#1}
   \Adv
143   \ifnum#1=1
145     \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.15)
      {\trebleclef};
147   \fi \ifnum#1=2
      \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.35)
149     {\trebleclef};
   \fi \ifnum#1=3
151     \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.8)
      {{\resizebox{0.7cm}{0.8cm}{\bassclef}}};
153   \fi \ifnum#1=4
      \node at (\theMxpos*0.5+0.1,-\theMypos*2+1)
155     {{\resizebox{0.7cm}{0.8cm}{\bassclef}}};
   \fi \ifnum#1=11
157     \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.4)
      {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
159   \fi \ifnum#1=12
      \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.6)
161     {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
   \fi \ifnum#1=13
163     \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.8)
      {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
165   \fi \ifnum#1=14
      \node at (\theMxpos*0.5+0.1,-\theMypos*2+1)
167     {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
   \fi \ifnum#1=15
169     \node at (\theMxpos*0.5+0.1,-\theMypos*2+1.2)
      {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
171   \fi
      \addtocounter{Mxpos}{2}
173 }

\newlength{\MWidth}
\newlength{\MHeight}
177 \newcommand{\MResize}[1]{
   \settowidth{\MWidth}{#1}
179   \settoheight{\MHeight}{#1}
   \resizebox{1.5\MWidth}{1.5\MHeight}{#1}
181 }
\newcommand{\TimeSig}[3][0]{
183   \Adv
   \ifnum#1=0
185   \node at (\theMxpos*0.5,-\theMypos*2+0.6) {\bf \MResize{#2}};

```

```

\newcommand{\Mxpos*0.5,-\Mypos*2+0.2}{\bf \MResize{#3}};
\fi\ifnum#1=1
\newcommand{\Mxpos*0.5,-\Mypos*2+0.5}{\meterC};
\fi\ifnum#1=2
\newcommand{\Mxpos*0.5,-\Mypos*2+0.575}{\allabreve};
\fi
\addtocounter{Mxpos}{3}
}

\newcommand{\Dot}[2][1]{
\foreach \d in {1,...,#1}
{\node at (\Mxpos*0.5+0.1+0.15*\d,-\Mypos*2-0.21+0.1*#2)
{\resizebox{0.1cm}{0.1cm}{\bullet}};}}
}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

\newcommand{\Head}[1]{
\node [opacity=0.0] (tmp) at (\Mxpos*0.5,-\Mypos*2-0.2+#1*0.1) {};
\fill [rotate around={16:(tmp.center)}] (\Mxpos*0.5-0.01,-\Mypos*2-0.195+#1*0.1) ellipse (4pt and 2.75pt);}

\newcommand{\HollowHead}[1]{
\node [opacity=0.0] (tmp) at (\Mxpos*0.5,-\Mypos*2-0.2+#1*0.1) {};
\fill [black,rotate around={16:(tmp.center)}] (\Mxpos*0.5-0.01,-\Mypos*2-0.195+#1*0.1) ellipse (4pt and 2.75pt);
\fill [white,rotate around={19:(tmp.center)}] (\Mxpos*0.5-0.01,-\Mypos*2-0.195+#1*0.1) ellipse (3.2pt and 1.35pt);}
}

\newcommand{\Enlarge}[3][0]{
\ifswap\Mswap\fi
\ifnoswap
\draw[line width=0.25mm] (\Mxpos*0.5+0.115,-\Mypos*2+#2*0.1-0.2) --
(\Mxpos*0.5+0.115,-\Mypos*2+#3*0.1+#1);
\else
\draw[line width=0.25mm] (\Mxpos*0.5-0.14,-\Mypos*2+#2*0.1-0.2) --
(\Mxpos*0.5-0.14,-\Mypos*2+#3*0.1-0.5-#1);
\fi
\ifswap\Mswap\fi
}

\newcommand{\Ledger}[1]{
\ifnum1>#1
\foreach \i in {0,...,#1} {
\ifthenelse{\isodd{\i}}{
\draw (\Mxpos*0.5-0.25,-\Mypos*2-0.2+\i*0.1) -- (\Mxpos*0.5+0.25,-\Mypos*2-0.2+\i*0.1);}
}
\fi\ifnum1<#1
\foreach \i in {12,...,#1} {
\ifthenelse{\isodd{\i}}{
\draw (\Mxpos*0.5-0.25,-\Mypos*2-0.2+\i*0.1) -- (\Mxpos*0.5+0.25,-\Mypos*2-0.2+\i*0.1);}
}
\fi
}

\newcommand{\Beam}[4]{
\ifswap\Mswap\fi
\ifnoswap

```

```

243     \draw[line width=1mm] (#1*0.5+0.103,-\theMypos*2+#2*0.1#4) -- (\theMxpos*0.5-0.372,-\theMypos*2+#3*0.1#4)
244     \else
245     \draw[line width=1mm] (#1*0.5-0.153,-\theMypos*2+#2*0.1-0.5-#4) -- (\theMxpos*0.5-0.625,-\theMypos*2+#3*0.1-0.5-#4)
246     \fi
247 \ifswap\Mswap\fi
248 }
249 \newcommand{\BeamS}[1]{
250 \ifswap\Mswap\fi
251 \ifnoswap
252     \ifnum\theMfirst=\theMmin
253     \begin{scope}
254         \clip (\theMxpos*0.5-0.4,-\theMypos*2+\theMmin*0.1-0.3) rectangle (\theMxpos*0.5+0.125,-\theMypos*2+0.05+\theMmin*0.1-0.3)
255         \draw[line width=1mm] (\theMxpos*0.5-0.411,-\theMypos*2+\theMmin*0.1#1) -- (\theMxpos*0.5+0.147,-\theMypos*2+\theMmin*0.1#1)
256     \end{scope}
257     \else
258     \begin{scope}
259         \clip (\theMxpos*0.5-0.4,-\theMypos*2+0.05+\theMmax*0.1#1) rectangle (\theMxpos*0.5+0.125,-\theMypos*2+0.05+\theMmax*0.1#1)
260         \draw[line width=1mm] (\theMxpos*0.5-0.411,-\theMypos*2+\theMmax*0.1#1) -- (\theMxpos*0.5+0.147,-\theMypos*2+\theMmax*0.1#1)
261     \end{scope}
262     \fi
263 \else
264     \ifnum\theMfirst=\theMmin
265     \begin{scope}
266         \clip (\theMxpos*0.5-0.645,-\theMypos*2+\theMmin*0.1-0.5-#1) rectangle (\theMxpos*0.5-0.129,-\theMypos*2+\theMmin*0.1-0.5-#1)
267         \draw[line width=1mm] (\theMxpos*0.5-0.7,-\theMypos*2+\theMmin*0.1-#1-0.48) -- (\theMxpos*0.5-0.1,-\theMypos*2+\theMmin*0.1-#1-0.48)
268     \end{scope}
269     \else
270     \begin{scope}
271         \clip (\theMxpos*0.5-0.65,-\theMypos*2+\theMmin*0.1-0.5-#1) rectangle (\theMxpos*0.5-0.129,-\theMypos*2+\theMmin*0.1-0.5-#1)
272         \draw[line width=1mm] (\theMxpos*0.5-0.7,-\theMypos*2+\theMmax*0.1-#1-0.48) -- (\theMxpos*0.5-0.1,-\theMypos*2+\theMmax*0.1-#1-0.48)
273     \end{scope}
274     \fi
275 \fi
276 \ifswap\Mswap\fi
277 }
278 \newcommand{\Semibreve}[1]{
279 \node at (\theMxpos*0.5,-\theMypos*2-0.2+#1*0.1) {\resizebox{0.3cm}{0.2cm}{\Ganz}};
280 }
281 \newcommand{\Minim}[1]{
282 \HollowHead{#1}
283 \ifnum#1<6\else
284 \Mswap
285 \fi
286 \Enlarge[0.3]{#1}{#1}
287 }
288 \newcommand{\Crotchet}[1]{
289 \Head{#1}
290 \ifnum#1<6\else
291 \Mswap
292 \fi
293 \Enlarge[0.3]{#1}{#1}
294 }
295 \newcommand{\Flag}[2][0]{
296 \ifswap\Mswap\fi
297 \ifnoswap

```

```

\begin{scope}
301 \clip (\theMxpos*0.5+0.13,-\theMypos*2+#2*0.1-0.75+#1) rectangle (\theMxpos*0.5+0.35,-\theMypos*2+#2*0.1-0.75+#1);
\draw (\theMxpos*0.5+0.05,-\theMypos*2-0.3+#2*0.1+#1) {\resizebox{0.55cm}{0.65cm}{\Acht}};
303 \end{scope}
\fill [white] (\theMxpos*0.5+0.205,-\theMypos*2-0.308+#2*0.1+#1) circle (1.75pt);
305 \else
\begin{scope}
307 \node [opacity=0.0] (tmp) at (\theMxpos*0.5,-\theMypos*2+#2*0.1) {};
\clip (\theMxpos*0.5-0.125,-\theMypos*2+#2*0.1-0.55+#1) rectangle (\theMxpos*0.5+0.09,-\theMypos*2+#2*0.1-0.55+#1);
309 \node[rotate around={180:(tmp.center)}] at (\theMxpos*0.5+0.175,-\theMypos*2+0.165+#2*0.1+#1) {\reflectbox{\Acht}};
\end{scope}
311 \fill [white] (\theMxpos*0.5-0.035,-\theMypos*2-0.161+#2*0.1-#1) circle (2pt);
\fi
313 \ifswap\Mswap\fi
}
315 \newcommand{\Quaver}[1]{
317 \Head{#1}
\ifnum#1<6\else
319 \Mswap
\fi
321 \Enlarge[0.4]{#1}{#1}
\Flag[0.4]{#1}
323 }
\newcommand{\Semiquaver}[1]{
325 \Head{#1}
\ifnum#1<6\else
327 \Mswap
\fi
329 \Enlarge[0.5]{#1}{#1}
\Flag[0.5]{#1}
331 \Flag[0.35]{#1}
}
\newcommand{\Demisemiquaver}[1]{
333 \Head{#1}
\ifnum#1<6\else
335 \Mswap
\fi
337 \Enlarge[0.6]{#1}{#1}
\Flag[0.6]{#1}
339 \Flag[0.45]{#1}
\Flag[0.3]{#1}
341 }
}
343 \newcommand{\Note}[2]{
\Adv\noswaptrue
345
\ifnum#1=1
347 \Semibreve{#2}
\fi
\ifnum#1=2
349 \Minim{#2}
\fi
\ifnum#1=4
351 \Crotchet{#2}
\fi
\ifnum#1=8
353 \Quaver{#2}
\fi
\ifnum#1=16
355 \Semiquaver{#2}
\fi
\ifnum#1=32

```

```

357     \Demisemiquaver{#2}
359     \fi
361     \Ledger{#2}
363     \addtocounter{Mxpos}{2}
365     \swapfalse
367 }
369
371 \newcommand{\PolyNote}[2]{
372     \Adv
373     \setcounter{Mfirst}{-50}
374     \setcounter{Mmax}{-50}
375     \setcounter{Mmin}{50}
376     \setcounter{Maux}{\theMxpos}
377
378     \ifnum#1=1
379     \foreach\n in {#2} {
380         \Semibreve{\n}\stepcounter{Mxpos}}
381     \fi\ifnum#1=2
382     \foreach\n in {#2} {
383         \Minim{\n}\stepcounter{Mxpos}}
384     \fi\ifnum#1=4
385     \foreach\n in {#2} {
386         \Crotchet{\n}\stepcounter{Mxpos}}
387     \fi\ifnum#1>4
388     \foreach\n in {#2} {
389         \Head{\n}\stepcounter{Mxpos}
390         \MMax{\n}\MMin{\n} % max, min, first
391         \ifnum\theMfirst=-50\setcounter{Mfirst}{\n}\fi
392     }
393     \noswaptrue
394     \ifnum\theMmax<6\else
395     \noswapfalse\fi
396     \setcounter{Mevaluate}{\theMxpos}\addtocounter{Mevaluate}{-\theMaux}
397     \setcounter{Mxpos}{\theMaux}
398     \ifnum\theMevaluate=2 % only two notes
399     \foreach\n in {#2} {
400         \ifnum#1=8
401         \Enlarge[0.4]{\n}{\n}
402         \fi\ifnum#1=16
403         \Enlarge[0.5]{\n}{\n}
404         \fi\ifnum#1=32
405         \Enlarge[0.6]{\n}{\n}
406         \fi
407         \stepcounter{Mxpos}
408     }
409     \addtocounter{Mxpos}{-1}
410     \ifnum#1=8
411     \BeamS{+0.4}
412     \fi\ifnum#1=16
413     \BeamS{+0.5}
414     \BeamS{+0.35}
415     \fi\ifnum#1=32
416     \BeamS{+0.6}
417     \BeamS{+0.45}
418     \BeamS{+0.3}
419     \fi

```



```

\else
415   \ifswap\Mswap\fi
      \ifnoswap
417   \setcounter{Mevaluate}{\theMmax}
      \else
419   \setcounter{Mevaluate}{\theMmin}
      \fi\ifswap\Mswap\fi
421   \foreach\n in {#2} {
      \ifnum#1=8
423     \Enlarge[0.4]{\n}{\theMevaluate}
      \fi\ifnum#1=16
425     \Enlarge[0.5]{\n}{\theMevaluate}
      \fi\ifnum#1=32
427     \Enlarge[0.6]{\n}{\theMevaluate}
      \fi
429     \stepcounter{Mxpos}
    }
431   \ifnum#1=8
      \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.4}
433   \fi\ifnum#1=16
      \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.5}
435   \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.35}
      \fi\ifnum#1=32
437   \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.6}
      \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.45}
439   \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.3}
      \fi
441   \fi
\fi
443   \setcounter{Mxpos}{\theMaux}
      \foreach\n in {#2} {
445       \Ledger{\n}\stepcounter{Mxpos}}
      \addtocounter{Mxpos}{1}
447   \swapfalse
}
449
\newcommand{\Rest}[1]{
451   \Adv

453   \ifnum#1=1
      \node at (\theMxpos*0.5,-\theMypos*2+0.52) {\resizebox{0.3cm}{0.15cm}{\GaPa}};
455   \fi\ifnum#1=2
      \node at (\theMxpos*0.5,-\theMypos*2+0.42) {\resizebox{0.3cm}{0.15cm}{\HaPa}};
457   \fi\ifnum#1=4
      \node at (\theMxpos*0.5,-\theMypos*2+0.4) {\resizebox{0.3cm}{0.7cm}{\ViPa}};
459   \fi\ifnum#1=8
      \node at (\theMxpos*0.5,-\theMypos*2+0.32) {\resizebox{0.3cm}{0.45cm}{\AcPa}};
461   \fi\ifnum#1=16
      \node at (\theMxpos*0.5,-\theMypos*2+0.3) {\resizebox{0.35cm}{0.525cm}{\SePa}};
463   \fi\ifnum#1=32
      \node at (\theMxpos*0.5,-\theMypos*2+0.375) {\resizebox{0.4cm}{0.6cm}{\ZwPa}};
465   \fi

467   \addtocounter{Mxpos}{2}
}
469
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

```

```

471 \newcommand{\Acc}[3][−0.25]{
473   \ifnum#2=0
475     \node at (\theMxpos*0.5#1,−\theMypos*2−0.2+0.1*#3)
       {\resizebox{0.15cm}{0.35cm}{\natural}};
477   \fi \ifnum#2=1
       \node at (\theMxpos*0.5#1,−\theMypos*2−0.2+0.1*#3)
       {\resizebox{0.15cm}{0.35cm}{\sharp}};
479   \fi \ifnum#2=−1
       \node at (\theMxpos*0.5#1,−\theMypos*2−0.1+0.1*#3)
       {\resizebox{0.15cm}{0.35cm}{\flat}};
481   \fi \ifnum#2=2
       \node at (\theMxpos*0.5#1,−\theMypos*2−0.205+0.1*#3)
       {\resizebox{0.15cm}{0.15cm}{\doubleslash}};
483   \fi \ifnum#2=−2
       \node at (\theMxpos*0.5#1,−\theMypos*2−0.1+0.1*#3)
       {\resizebox{0.2cm}{0.35cm}{\doubleflat}};
485   \fi
487 }
489 }

491 \newcounter{Maux}
493 \newcommand{\AccFN}[2][−0.25]{
495   \KB{#2}
497   \Acc[#1]{0}{\theMaux}
499 \newcommand{\AccSN}[2][−0.25]{
501   \KS{#2}
503   \Acc[#1]{0}{\theMaux}

505 \newcommand{\KB}[1]{
507   \setcounter{Maux}{#1}
       \addtocounter{Maux}{−3}

509   \ifnum\theMclef=1
       \addtocounter{Maux}{−2}
511   \fi \ifnum\theMclef=2
       \addtocounter{Maux}{0}
513   \fi \ifnum\theMclef=3
       \addtocounter{Maux}{3}
515   \fi \ifnum\theMclef=4
       \addtocounter{Maux}{−2}
517   \fi \ifnum\theMclef=11
       \addtocounter{Maux}{2}
519   \fi \ifnum\theMclef=12
       \addtocounter{Maux}{−3}
521   \fi \ifnum\theMclef=13
       \addtocounter{Maux}{−1}
523   \fi \ifnum\theMclef=14
       \addtocounter{Maux}{1}
525   \fi \ifnum\theMclef=15
       \addtocounter{Maux}{3}
527   \fi

```

```

529     \MModulo{\theMaux}{7}
        \setcounter{Maux}{\theMModulo}
        \addtocounter{Maux}{3}
531   }
    \newcommand{\KS}[1]{
533     \setcounter{Maux}{#1}
        \addtocounter{Maux}{-5}
535
        \ifnum\theMclef=1
537         \addtocounter{Maux}{-2}
        \fi \ifnum\theMclef=2
539         \addtocounter{Maux}{0}
        \fi \ifnum\theMclef=3
541         \addtocounter{Maux}{3}
        \fi \ifnum\theMclef=4
543         \addtocounter{Maux}{-2}
        \fi \ifnum\theMclef=11
545         \addtocounter{Maux}{2}
        \fi \ifnum\theMclef=12
547         \addtocounter{Maux}{-3}
        \fi \ifnum\theMclef=13
549         \addtocounter{Maux}{-1}
        \fi \ifnum\theMclef=14
551         \addtocounter{Maux}{1}
        \fi \ifnum\theMclef=15
553         \addtocounter{Maux}{3}
        \fi
555     \MModulo{\theMaux}{7}
        \setcounter{Maux}{\theMModulo}
557     \addtocounter{Maux}{5}
    }
559
    \newcommand{\Key}[1]{
561     \setcounter{Mkey}{#1}
        \ifnum#1=-1
563         \AccF[-0.2]{6}
        \fi \ifnum#1=-2
565         \AccF[-0.35]{6}
        \AccF[-0.05]{9}
567         \fi \ifnum#1=-3
        \AccF[-0.4]{6}
569         \AccF[-0.15]{9}
        \AccF[-0.0]{5}
571         \fi \ifnum#1=-4
        \AccF[-0.45]{6}
573         \AccF[-0.25]{9}
        \AccF[-0.1]{5}
575         \AccF[+0.1]{8}
        \fi \ifnum#1=-5
577         \AccF[-0.45]{6}
        \AccF[-0.25]{9}
579         \AccF[-0.15]{5}
        \AccF[+0.05]{8}
581         \AccF[+0.15]{4}
        \fi \ifnum#1=-6
583         \AccF[-0.45]{6}
        \AccF[-0.3]{9}

```

```

585     \AccF[-0.2]{5}
        \AccF[-0.05]{8}
587     \AccF[+0.05]{4}
        \AccF[+0.2]{7}
589     \fi \ifnum#1=-7
        \AccF[-0.45]{6}
591     \AccF[-0.35]{9}
        \AccF[-0.25]{5}
593     \AccF[-0.15]{8}
        \AccF[-0.05]{4}
595     \AccF[+0.05]{7}
        \AccF[+0.15]{3}
597     \fi
        \ifnum#1=1
599     \AccS[-0.2]{10}
        \fi \ifnum#1=2
601     \AccS[-0.35]{10}
        \AccS[-0.05]{7}
603     \fi \ifnum#1=3
        \AccS[-0.425]{10}
605     \AccS[-0.2]{7}
        \AccS[-0.025]{11}
607     \fi \ifnum#1=4
        \AccS[-0.45]{10}
609     \AccS[-0.25]{7}
        \AccS[-0.1]{11}
611     \AccS[+0.1]{8}
        \fi \ifnum#1=5
613     \AccS[-0.45]{10}
        \AccS[-0.3]{7}
615     \AccS[-0.15]{11}
        \AccS[-0.0]{8}
617     \AccS[+0.15]{5}
        \fi \ifnum#1=6
619     \AccS[-0.4]{10}
        \AccS[-0.3]{7}
621     \AccS[-0.2]{11}
        \AccS[-0.1]{8}
623     \AccS[+0.0]{5}
        \AccS[+0.1]{9}
625     \fi \ifnum#1=7
        \AccS[-0.45]{10}
627     \AccS[-0.35]{7}
        \AccS[-0.25]{11}
629     \AccS[-0.15]{8}
        \AccS[-0.05]{5}
631     \AccS[+0.05]{9}
        \AccS[+0.15]{6}
633     \fi
        \stepcounter{Mxpos}
635 }

637 \newcommand{\NoKey}[1]{
        \setcounter{Mkey}{0}
639     \ifnum#1=-1
        \AccFN[-0.2]{6}
641     \fi \ifnum#1=-2

```

643        \AccFN[-0.35]{6}  
           \AccFN[-0.05]{9}  
           \fi \ifnum#1=-3  
 645        \AccFN[-0.4]{6}  
           \AccFN[-0.15]{9}  
 647        \AccFN[-0.0]{5}  
           \fi \ifnum#1=-4  
 649        \AccFN[-0.45]{6}  
           \AccFN[-0.25]{9}  
 651        \AccFN[-0.1]{5}  
           \AccFN[+0.1]{8}  
 653        \fi \ifnum#1=-5  
           \AccFN[-0.45]{6}  
 655        \AccFN[-0.25]{9}  
           \AccFN[-0.15]{5}  
 657        \AccFN[+0.05]{8}  
           \AccFN[+0.15]{4}  
 659        \fi \ifnum#1=-6  
           \AccFN[-0.45]{6}  
 661        \AccFN[-0.3]{9}  
           \AccFN[-0.2]{5}  
 663        \AccFN[-0.05]{8}  
           \AccFN[+0.05]{4}  
 665        \AccFN[+0.2]{7}  
           \fi \ifnum#1=-7  
 667        \AccFN[-0.45]{6}  
           \AccFN[-0.35]{9}  
 669        \AccFN[-0.25]{5}  
           \AccFN[-0.15]{8}  
 671        \AccFN[-0.05]{4}  
           \AccFN[+0.05]{7}  
 673        \AccFN[+0.15]{3}  
           \fi  
 675        \ifnum#1=1  
           \AccSN[-0.2]{10}  
 677        \fi \ifnum#1=2  
           \AccSN[-0.35]{10}  
 679        \AccSN[-0.05]{7}  
           \fi \ifnum#1=3  
 681        \AccSN[-0.425]{10}  
           \AccSN[-0.2]{7}  
 683        \AccSN[-0.025]{11}  
           \fi \ifnum#1=4  
 685        \AccSN[-0.45]{10}  
           \AccSN[-0.25]{7}  
 687        \AccSN[-0.1]{11}  
           \AccSN[+0.1]{8}  
 689        \fi \ifnum#1=5  
           \AccSN[-0.45]{10}  
 691        \AccSN[-0.3]{7}  
           \AccSN[-0.15]{11}  
 693        \AccSN[-0.0]{8}  
           \AccSN[+0.15]{5}  
 695        \fi \ifnum#1=6  
           \AccSN[-0.4]{10}  
 697        \AccSN[-0.3]{7}  
           \AccSN[-0.2]{11}

```

699     \AccSN[-0.1]{8}
        \AccSN[+0.0]{5}
701     \AccSN[+0.1]{9}
        \fi \ifnum#1=7
703     \AccSN[-0.45]{10}
        \AccSN[-0.35]{7}
705     \AccSN[-0.25]{11}
        \AccSN[-0.15]{8}
707     \AccSN[-0.05]{5}
        \AccSN[+0.05]{9}
709     \AccSN[+0.15]{6}
        \fi
711     \stepcounter{Mxpos}
    }
713
715     %%%%%%%%%%%
    \newcommand{\Text}[2][0]{
717     \node at (\theMxpos*0.5,-\theMypos*2-0.7+#1) {\#2};
    }
719
    \newcommand{\Chord}[1][1]{
721     \addtocounter{Mxpos}{-1}
        \addtocounter{Mxpos}{-#1}
723     \newcommand{\Space}[1][2]{\Adv\addtocounter{Mxpos}{#1}}

725     \newcommand{\Bar}{
        \ifnum\theMxpos=23
727         \addtocounter{Mxpos}{-1}
        \fi
729         \draw (\theMxpos*0.5,-\theMypos*2+0.8) -- (\theMxpos*0.5,-\theMypos*2);
        \addtocounter{Mxpos}{2}
731
    \newcommand{\DoubleBar}{
733         \ifnum\theMxpos=23
            \addtocounter{Mxpos}{-1}
735         \fi
            \draw (\theMxpos*0.5-0.06,-\theMypos*2+0.8) -- (\theMxpos*0.5-0.06,-\theMypos*2);
737            \draw (\theMxpos*0.5+0.06,-\theMypos*2+0.8) -- (\theMxpos*0.5+0.06,-\theMypos*2);
            \addtocounter{Mxpos}{2}
739
    \newcommand{\FinalBar}{
741         \ifnum\theMxpos=23
            \addtocounter{Mxpos}{-1}
743         \fi
            \draw (\theMxpos*0.5-0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5-0.12,-\theMypos*2);
745            \draw[line width=1mm] (\theMxpos*0.5,-\theMypos*2+0.805) -- (\theMxpos*0.5,-\theMypos*2-0.005);
            \addtocounter{Mxpos}{2}
747     }

749     \newcommand{\MString}[3]{
        \ifthenelse{\equal{#1}{#2}}
751         {\#3}
        {}
753     }

755     \newcommand{\RepeatBar}[1]{

```

```

\ifnum\theMxpos=23
757   \addtocounter{Mxpos}{-1}
\fi
759   \MString{#1}{L}{
\draw[draw[|line width=1mm] (\theMxpos*0.5-0.24,0.4-\theMypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
761   \draw (\theMxpos*0.5-0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5-0.12,-\theMypos*2);
\draw[|line width=1mm] (\theMxpos*0.5,-\theMypos*2+0.805) -- (\theMxpos*0.5,-\theMypos*2-0.005);
763   }
\MString{#1}{R}{
765   \draw[|line width=1mm] (\theMxpos*0.5,-\theMypos*2+0.805) -- (\theMxpos*0.5,-\theMypos*2-0.005);
\draw (\theMxpos*0.5+0.24,0.4-\theMypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
767   \draw (\theMxpos*0.5+0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5+0.12,-\theMypos*2);
}
769   \MString{#1}{LR}{
\draw[draw[|line width=1mm] (\theMxpos*0.5-0.24,0.4-\theMypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
771   \draw (\theMxpos*0.5-0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5-0.12,-\theMypos*2);
\draw[|line width=1mm] (\theMxpos*0.5,-\theMypos*2+0.805) -- (\theMxpos*0.5,-\theMypos*2-0.005);
773   \draw (\theMxpos*0.5+0.24,0.4-\theMypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
\draw (\theMxpos*0.5+0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5+0.12,-\theMypos*2);
775   }
\addtocounter{Mxpos}{2}
777 }

779 \newcommand{\Adv}{
\ifnum\theMxpos>20
781   \addtocounter{Mypos}{1}
\setcounter{Mxpos}{0}
783   \ifnum\theMclef=0\else
\Clef{\theMclef}
785   \fi
\ifnum\theMkey=0\else
787   \Key{\theMkey}
\fi
789 }
\fi

791 \newcounter{Mtied}
793 \newcounter{MtiedA}
\newcounter{MtiedB}
795 \newenvironment{tied}[2]
{\setcounter{Mtied}{\theMxpos}\setcounter{MtiedA}{#1}\setcounter{MtiedB}{#2}}
797 {\draw (\theMtied*0.5,-\theMypos*2+0.1*\theMtiedA) to[out=45,in=135] (-1+\theMxpos*0.5,-\theMypos*2+0.1*\theMtiedB);

799 \newcounter{Mslurred}
801 \newcounter{MslurredA}
\newcounter{MslurredB}
803 \newenvironment{slurred}[2]
{\setcounter{Mslurred}{\theMxpos}\setcounter{MslurredA}{#1}\setcounter{MslurredB}{#2}}
805 {\draw (\theMslurred*0.5,-\theMypos*2+0.1*\theMslurredA-0.35) to[out=-45,in=-135] (-1+\theMxpos*0.5,-\theMypos*2+0.

```